

Response to Review: JCGS-25-420.R1

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We thank the anonymous associate editor and reviewers for their careful reviews. What follows is our point-by-point response to reviewer comments. *Our responses are in italic text.*

This revision and the comments given by the authors do not successfully address most of my previous major and minor concerns. My impression is that they attempted a “quick fix” by adding some new material here and there without substantially reworking the narrative, evaluation, and presentation quality of the manuscript and did not engage with the substance of my previous points in good faith.

Despite my criticisms, I do think the core idea of HBE + tour overlays is very interesting and innovative and has large potential value as a diagnostic tool for NLDR embeddings. However, in its current form, the manuscript is not suitable for publication in my opinion.

Thank you. In response to the reviewer feedback, we have rearranged the paper substantially. We believe that it is more clearly focused and concise, and addresses all of the reviewer concerns. This also means that although we have responded to review point-by-point response, the rearrangement may make the changes relative to the original harder to see. So we have included a full diff of the versions, so the key changes might be clearer.

Prior Major Points

1) Missing comparisons to standard embedding-quality measures (co-ranking / RNX, trustworthiness/continuity, stress, Shepard diagrams, etc.)

Partly addressed. The revision now explicitly discusses several established metrics in Section 3.4, and it includes numerical/visual comparisons of HBE against metrics such as “rARNX”, “rRTA”, “rGS”, and a Shepard-diagram-based correlation (e.g., Figure 8 and the surrounding discussion). However, the new material mostly illustrates that metrics can disagree rather than establishing when/why HBE should be preferred or how an analyst should reconcile HBE with

existing (and complementary) local-vs-global criteria. The revised narrative is too dismissive of disagreement between metrics as “confusion” rather than treating it as an unavoidable multi-objective trade-off (see below).

We appreciate this point. Changes are explained in the review comments for the second review below.

2) Overselling the method as a decision aid for choosing “the best” 2-D embedding vs positioning it primarily as a diagnostic/exploratory tool

Not addressed (and arguably worse). The paper still contains a full section titled “Choosing the best 2-D layout” (Section 4), and the surrounding prose continues to frame the workflow as “select the most accurate representation” rather than “diagnose and understand distortions and failure modes of a chosen representation.” In their response letter, the authors themselves explicitly stated that HBE is intended as a diagnostic tool rather than a definitive selection metric” but the submitted manuscript text does not reflect this position at all.

Noted. Changes explained in the review comments for the second review below.

3) Section 3.5 (out-of-sample / prediction into an embedding)

Only very partially addressed. Per the response letter, citations to related work have been added and the authors frame this as future work. But the manuscript still gives it prominent space in the methods section; there is no theoretical argument, no empirical validation, and no comparison to existing out-of-sample embedding approaches.

Changes explained in the review comments for the second review below.

4) Section 3.6 (parameter tuning guidance)

Not addressed in substance. The section is longer, but still reads like: “Try many settings, draw many plots, then pick something that looks OK.” I still do not see clear, reproducible guidance that an analyst can follow to choose binwidth etc parameters in a systematic way. If none can be provided, this should be clearly stated and the method reframed as inherently exploratory/diagnostic rather than selection-oriented.

Point taken. This is more explicit in the revision.

5) Writing/proofreading and overall presentation quality

Not addressed. Despite the claim in the response letter that the manuscript was carefully proofread, the revision still contains a distracting density of typos, malformed sentences, inconsistent notation, and colloquial phrasing (examples below from Section 3.4 alone). The writing quality is still far below what I would expect for a publication-ready manuscript in JCGS.

This is not a fair statement, but we take it in stride. We have carefully proofed the revision.

Logistical aspects of the revision

The revision process is made unnecessarily difficult because none of the changes are highlighted and the response letter does not point to precise locations (page/line) nor quotes key replacements. I find this rather inconsiderate and fairly unprofessional. A tracked-changes PDF (or a detailed change log keyed to reviewer comments) would substantially reduce friction for reviewers and editors if another revision is submitted.

We have submitted a diff file between versions that will help to see the key changes made in the revision.

Summary

IMO, the manuscript still has at least two critical problems:

1. Claim–evidence mismatch: It is still written as if it provides a reasonably objective procedure for choosing the best 2-D layout, but the evaluation is not strong enough to support that framing.

Addressed below.

2. Presentation quality: The prose is sloppy enough (typos, malformed sentences, non standard/unmotivated notation) that it materially impairs readability and confidence.

Addressed above.

Major Issues

1. Still oversells “Choosing the best 2-D layout”

Section 4 is still titled “**Choosing the best 2-D layout**”, and the text repeatedly suggests that the goal is to select an “accurate” representation over a pretty but wrong representation. This framing remains problematic for the basic reason I raised previously: in 2-D NLDR,

there is generally no single optimum – there is a trade-off between local and global structure preservation, and “best” is always implicitly “best for some downstream purpose.”

The revised section better explains the use of HBE relative to other measures. The other measures are shown and compared for the simulated example and the applications.

The section on choosing layouts better explains how to use HBE. It should be considered a tool to help select a layout from a range of choices, not just as a diagnostic tool.

2. Validation remains insufficient for a selection claim (even with new metric comparisons)

The added comparisons to rARNX/rRTA/rGS/rSC are useful, but they do not resolve the central question: **Does HBE reliably help choose better embeddings, and in what sense of “better”?**

Right now, the paper still reads like a sequence of worked examples. The new metric comparison section mainly shows that different metrics rank layouts differently and then argues (qualitatively) that some metrics overemphasize global separation. That may well be true in specific clustering-heavy scenarios, but it does not constitute validation of HBE as a general purpose criterion.

In particular, if the paper’s main claim is **diagnostics/exploration**, then the evidence should demonstrate that HBE + the tour overlay helps analysts localize + interpret distortions that are hard to see otherwise. (This seems to be there, already, but could be sharpened). If the paper’s claim is **selection**, then the evidence given in the paper must demonstrate that a well specified selection procedure based on HBE selects embeddings that are “better” according to well-defined criteria across a range of controlled scenarios. (This is currently missing and, IMO, would almost necessarily require a simulation study with data with various known high dimensional structures as ground truth.)

The paper does describe how to choose the best layout from a selection of layouts. The text has been revised to make this point clearer.

3. Section 3.6 still does not give reproducible, non-ad-hoc tuning guidance

With all due respect, my takeaway from the expanded Section 3.6 is still: “You can try a lot of things and draw some graphs and then maybe pick one of these settings, but who knows...” I might be missing something, but I do not see a clear recipe that an analyst could follow without substantial subjective judgment calls. If no such recipe can be given, this should be clearly stated and the method reframed as inherently exploratory/diagnostic rather than selection-oriented.

We have moved the section containing the discussion on tuning to the appendix. This is primarily for deciding on a binwidth to use to make the representation of the model in high dimensions. It is not relevant for choosing among a set of candidate layouts.

4. Section 3.5 (out-of-sample embedding) still reads like an unvalidated method

The response letter says this is “future work,” and some citations were added. That is helpful, but it certainly cannot be presented as if it were an alternative to established projection or out-of-sample embedding methods just by claiming that it is. Without at least a basic empirical check (e.g., does it preserve neighborhood structure for held-out points as well as established out-of-sample methods?), this section should not be presented as a usable alternative.

We have revised Section 3.5 (now 7) to better explain the approach. The primary comparison is with one existing approach available in UMAP. Our approach is very simple, but not entirely unlike the UMAP procedure. We show how our approach would provide reasonable predicted positions for each of the layouts shown in the simulated example. This demonstrates clearly why our approach is actually useful and could be used for any NLDR techniques where the full data is available. We do acknowledge your point that this has plenty of room for improvement, but argue that it should be part of potential future work.

5. Writing, proofreading, and notation quality are still far below publication-ready

The revised paper seems fairly sloppy and badly written overall. It was obviously not proofread or edited with any amount of proper diligence. There are too many typos, inappropriate colloquialisms, weird/bad phrasings, and non-sequiturs to list.

Just taking Section 3.4 as an example:

- **p. 12:** “ARNX” is not the abbreviation used in the cited work as far as I remember (and in any case should be carefully defined and kept consistent with the literature).

*The **ARNX** notation has been corrected and aligned with standard usage in the literature, and all metric names and abbreviations are now clearly defined and used consistently throughout the manuscript.*

- **p. 13, l. 6:** “The Shepard diagram (Shepard 1962) and its associated Spearman correlation (SC) (Spearman 1961) between p-D and k-D distances.” – not a sentence (no verb/object).

Sentence fragments and malformed constructions (such as the Shepard diagram description noted on p. 13) have been rewritten as complete, grammatically correct sentences.

- **p. 13:** “model here is similar to a confirmatory factor analysis model (see a general explanation in Brown (2015)), $T + \hat{E}$ ” – the notation $T + \hat{E}$ is not introduced nor explained anywhere, and not used again anywhere else either; it just confuses/distracts.

Fixed.

- **p. 13, l. 31:** “our approach is similar to the ones used in this area. it is based on” – capitalization error.

Capitalization, punctuation, and phrasing errors have been corrected throughout.

- **p. 13:** “these fitted values might also be denoted as $\hat{X}$ ” – notation introduced and then never used again anywhere throughout the text.

Notation fixed.

- **p. 14:** “which occurs due the intentional design that one cluster is slightly 3-D and perfectly captured by a 2-D layout” – not valid English, and also plausibly inverts the intended meaning. I assume this should be something like: “which occurs due to the intentional design that this cluster is slightly 3-D and thus not able to be perfectly represented/preserved by a 2-D layout.”

Checked.

A similar density of errors can be found in most other sections. To me, this still seems far from publication-ready in terms of writing quality (scientific merit and remaining open points of discussion notwithstanding).

Checked.